Fine-Grained Bird Recognition: The Role of Foreground and Background

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Abstract

We study fine-grained recognition of 200 bird species using the Caltech-UCSD Birds-200-2011 (CUB) dataset. The central question is how the input region and geometric preprocessing affect recognition: does preserving the complete annotated bird region improve accuracy, and can any gain be separated from changes in scale, cropping, and background? We fine-tune an ImageNet-pretrained ResNet-18 under a fixed train/validation split and compare three matched conditions: the full image, a 10%-expanded bird crop followed by center cropping, and a 10%-expanded bird crop resized with aspect ratio preserved and padded to a square. Across three training seeds, the letterbox condition reaches $74.31\% \pm 0.75\%$ validation Top-1 accuracy, compared with $70.31\% \pm 1.13\%$ for the full image and $67.44\% \pm 0.05\%$ for center crop. The official CUB test partition remains untouched. These results are preliminary. A local control that holds the letterbox geometry fixed obtains $74.00\% \pm 0.36\%$ with bird regions and $65.25\% \pm 0.44\%$ with whole images, so foreground localization or target scale is important in this pipeline, although the background effect is not isolated. The remaining work will inspect background perturbations and model behavior, then perform one final held-out test evaluation.

1. Introduction

Fine-grained visual classification requires distinguishing categories that share a global shape and differ in localized visual evidence, such as bill shape, head markings, and wing color. CUB-200-2011 provides an appropriate controlled setting: it contains 200 bird species, annotated boxes, image labels, and an official train/test division [1]. A high-capacity pre-trained convolutional network can model small visual differences, but its predictions may also depend on background scene cues or on image regions that are discarded by preprocessing.

Our project asks whether focusing computation on the annotated bird region helps a practical recognition pipeline, and whether the effect survives changes in input geometry. The working hypothesis is that preserving the complete bird region will reduce accidental removal of discriminative parts, but the benefit may depend on bird size and on the amount of retained context. We therefore treat the center-crop and letterbox conditions as preprocessing comparisons, not as evidence that background is inherently harmful.

The milestone contributes a reproducible baseline, a fixed validation protocol, and a three-seed comparison of three input conditions. It also identifies a confound in the current best result: the letterbox pipeline removes random resized cropping during training and center cropping during validation in addition to changing the bird region. The next phase is designed to separate these effects and then evaluate the selected model once on the official test set.

2. Problem Statement and Data

2.1. Dataset and splits

CUB-200-2011 contains 11,788 images from 200 classes, with 5,994 images in the official training partition and 5,794 in the official test partition. We keep the official test partition unopened while developing the method. From the official training partition, we construct a class-stratified validation split using seed 42: approximately 20% of each class is assigned to validation, producing 4,794 training and 1,200 validation images. All methods use the same split and the split hash is stored with every result.

Each image has a ground-truth bounding box. In the crop-aware conditions, the box is expanded by 10% of its width or height on each side, clipped at the image boundary, and then passed through the condition-specific geometric pipeline. This use of an exact box is an additional annotation requirement and must be stated explicitly when comparing against the unrestricted full-image setting.

2.2. Prediction target and metrics

The model predicts one of 200 species labels. We report Top-1 accuracy, Top-5 accuracy, and Macro-F1 on the fixed validation split. Top-1 measures the main decision, Top-5 tests whether the correct species remains among the leading candidates, and Macro-F1 gives each species equal weight rather than allowing frequent classes to dominate. Means and sample standard deviations summarize three training seeds (42, 43, and 44); they are not confidence intervals.

3. Technical Approach

3.1. Backbone and training protocol

We use ResNet-18 with ImageNet-pretrained weights [2] and replace its final fully connected layer with a 200-way classifier. Training uses ten epochs, batch size 32, AdamW, weight decay 10^{-4}, and mixed precision. For the first two epochs, the backbone is frozen and only the classifier is trained; all parameters are then fine-tuned with learning rates 10^{-3} for the classifier and 10^{-4} for the backbone. At each epoch we retain the checkpoint with the best validation Top-1 accuracy. The same backbone, loss, optimizer, epoch budget, and split are used across input conditions.

3.2. Input conditions

The three preliminary conditions are:

1. Full image. Training applies random resized cropping and horizontal flipping; validation applies resize-to-256 followed by a 224×224 center crop.

2. Bird center crop. The 10%-expanded bird box is extracted and then undergoes the same legacy training and validation geometry.

3. Bird letterbox. The 10%-expanded box is resized so that its longest side is 224 pixels, preserving aspect ratio, and is centered on a 224×224 canvas. The fill color is RGB $(124, 116, 104)$, close to the ImageNet normalization mean. Training keeps only a horizontal flip; validation is deterministic.

The first two conditions provide the previously established baseline and a crop control. The letterbox condition aims to preserve the complete bird while avoiding the head or tail truncation that can occur with center cropping. However, it also changes the training distribution because random resized cropping is removed. We therefore describe the current comparison as a pipeline-level result rather than a causal padding ablation.

Condition	Top-1 (%)	Top-5 (%)	Macro-F1 (%)
Full image	70.31 ± 1.13	91.42 ± 0.22	70.02 ± 0.92
Center crop	67.44 ± 0.05	89.64 ± 0.10	67.06 ± 0.23
Letterbox	74.31 ± 0.75	93.06 ± 0.42	73.97 ± 0.88

Table 1. Validation results over three training seeds. Values are mean ± sample standard deviation; the standard deviations are not confidence intervals. The official test set was not used.

3.3. Planned analyses

We have completed a whole-image letterbox control with the same reduced augmentation, separating geometric preservation from the use of a bird box. Next, we will apply controlled background perturbations (blur, occlusion, and background-only diagnostic inputs) and report accuracy changes by bird-size stratum. We will inspect Grad-CAM responses and confusion pairs [3], including cases where letterbox improves or worsens the prediction. Finally, after freezing the configuration, we will evaluate the selected checkpoint once on the official test partition.

4. Intermediate Results

Table 1 and Figure 1 summarize the preliminary comparison. Letterbox obtains the highest mean score on all three metrics and improves Top-1 by 4.00 percentage points over the full image and by 6.86 points over bird center crop. The direction is consistent across all three training seeds: paired Top-1 improvements over the full-image condition are 2.75, 4.92, and 4.33 points for seeds 42, 43, and 44, respectively. The lower within-condition variability of center crop should not be interpreted as greater generalization, because its mean is substantially lower.

To separate input content from augmentation geometry, whole-image and bird-region letterbox conditions were each retrained for three seeds on the same RTX 3060 Laptop GPU with 6 GB memory. The bird-region condition reaches 74.00% ± 0.36% Top-1, while the whole-image condition reaches 65.25% ± 0.44%. The paired gains are 9.33, 8.08, and 8.83 points for seeds 42, 43, and 44. This rules out removal of random cropping as the sole explanation, but changing the region also changes target scale and retained background.

The crop-only comparison was negative on average: center crop is 2.86 points below the full image. This result illustrates why region selection and geometric processing should be considered together. A narrow crop can remove context and can also cut off a head, tail, or other fine-grained evidence. In qualitative inspection,

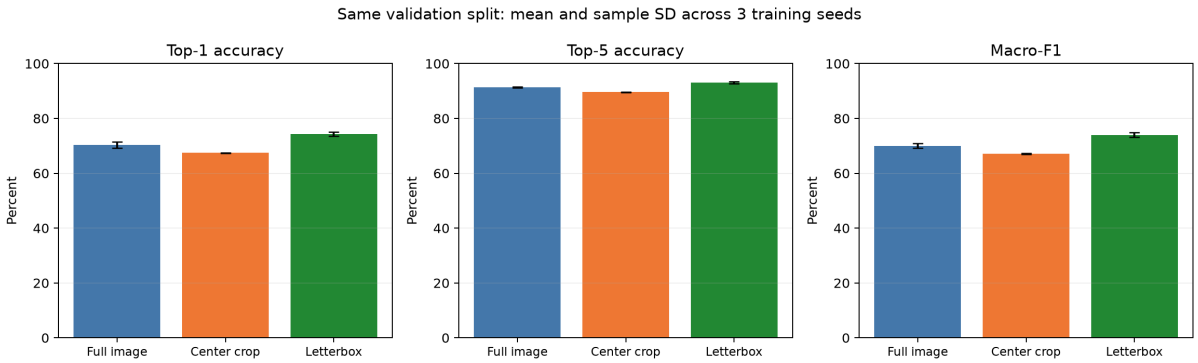


Figure 1. Same-validation-set comparison across training seeds 42–44. Error bars show sample standard deviation, not confidence intervals.

center-cropped inputs sometimes omitted visible parts of the bird, whereas the letterbox inputs retained the whole expanded region. These examples are selected for explanation and are not an estimate of the overall error distribution.

Bird-size stratification suggests that the benefit is not uniform. For images whose box occupies less than 10% of the image area ($n = 33$), the three-seed mean Top-1 is 61.62% for full image, 75.76% for center crop, and 82.83% for letterbox. The sample is small, so this is an exploratory observation rather than a precise subgroup estimate. For the 10–30% subgroup ($n = 539$), letterbox averages 73.59%, compared with 71.06% for full image and 65.99% for center crop.

5. Discussion and Risks

The current evidence supports continuing with the region-preserving letterbox pipeline as the provisional primary method. The whole-image letterbox control shows that the gain is not explained solely by removing random cropping, but the crop also increases effective bird scale and removes context. Background perturbation experiments will provide complementary evidence rather than definitive causal proof, because artificial occlusion also changes image statistics.

There are additional limitations. The pipeline assumes that a bird box is available; an unconstrained upload interface would require automatic localization or manual annotation. The CUB documentation notes possible image overlap with ImageNet, so ImageNet pretraining may expose the model to related images [1]. Three seeds provide preliminary stability evidence, but do not establish statistical significance or deterministic training. Finally, every reported number is validation performance; opening the test set before the final configuration is frozen would compromise its role as an independent evaluation.

6. Milestone Status and Next Steps

The data pipeline, model training, checkpointing, validation evaluation, and three-seed comparison are implemented. The project now has a validated baseline, a clear preprocessing comparison, and an explicit test-set policy. The immediate next steps are:

- separate the background and geometry effects with controlled perturbations;
- add Grad-CAM and confusion analysis to explain improvements and failures;
- freeze the final configuration, evaluate the official test partition once, and prepare the six-to-eight-page final report.

References

[1] C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie. The Caltech-UCSD Birds-200-2011 Dataset. CNS-TR-2011-001, 2011.

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